

Formal Invariant Verification & Proof Audit Certificate

Deterministic Cryptarithmic CSP Solution: SEND + MORE = MONEY

Reconciliation Engine (Autonomous Verification Pipeline COMPOSE-002)

September 2026

Certificate of Epistemic Invariant Verification

Certificate ID: CERT-COMPOSE-002-2026-09-15
Verification Tier: E3 - DETERMINISTICALLY VERIFIED
Verifier Engine: sim/reconciliation_engine.py (Zero-Token Invariant Auditor)
Solver Engine: AI-Constraint-Solver (Minimum Remaining Values Backtracking)

1. Problem Specification & Formal Constraints

The target combinatorial constraint satisfaction problem is defined over 8 unique alphabet symbols:

$$\text{SEND} + \text{MORE} = \text{MONEY}$$

Subject to the formal domain assertions:

$$\begin{aligned}\mathcal{D}(S), \mathcal{D}(M) &\subseteq \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ \mathcal{D}(E), \mathcal{D}(N), \mathcal{D}(D), \mathcal{D}(O), \mathcal{D}(R), \mathcal{D}(Y) &\subseteq \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ \text{AllDifferent}(S, E, N, D, M, O, R, Y)\end{aligned}$$

2. Solver Output & Assignment Vector

The backtracking search engine discovered the unique satisfying assignment in **12 ms** with **24 backtracks**:

Variable	Digit Assignment	Domain Valid	Non-Zero Lead Check
S	9	✓ ∈ [1, 9]	✓ $S \neq 0$
E	5	✓ ∈ [0, 9]	✓
N	6	✓ ∈ [0, 9]	✓
D	7	✓ ∈ [0, 9]	✓

Variable	Digit Assignment	Domain Valid	Non-Zero Lead Check
M	1	$\checkmark \in [1, 9]$	$\checkmark M \neq 0$
O	0	$\checkmark \in [0, 9]$	$\checkmark$
R	8	$\checkmark \in [0, 9]$	$\checkmark$
Y	2	$\checkmark \in [0, 9]$	$\checkmark$

3. Independent Mathematical Verification Audit

The `reconciliation_engine` independently evaluated the arithmetic sum without reliance on the solver's internal state:

$$\text{val}(\text{SEND}) = 9 \cdot 10^3 + 5 \cdot 10^2 + 6 \cdot 10^1 + 7 = \mathbf{9,567}$$

$$\text{val}(\text{MORE}) = 1 \cdot 10^3 + 0 \cdot 10^2 + 8 \cdot 10^1 + 5 = \mathbf{1,085}$$

$$\text{val}(\text{MONEY}) = 1 \cdot 10^4 + 0 \cdot 10^3 + 6 \cdot 10^2 + 5 \cdot 10^1 + 2 = \mathbf{10,652}$$

Direct Assertion Check:

$$9,567 + 1,085 = 10,652 \quad \Longleftrightarrow \quad \mathbf{10,652 == 10,652} \quad [\mathbf{VERIFIED: TRUE}]$$

$$\text{DistinctDigits}(\{9, 5, 6, 7, 1, 0, 8, 2\}) = 8 \quad [\mathbf{VERIFIED: TRUE}]$$

4. Invariant Assertion Status

Formal Verification Attestation

STATUS: ZERO DISCREPANCY VERIFIED

The assignment satisfies all algebraic and uniqueness invariants. Verified deterministically by independent AST arithmetic audit. Zero human intervention or external token cost.